Group Assignment – Conducting a Code Review

Each student in their group must create a Pull Request (PR) from their own branch in their individual garage repository, and then have the group members review it in GitHub.

The purpose of this exercise is to practise reading code, giving constructive feedback, and using the Git workflow—not just finding bugs.

Preparation

Each student has their own private repository on their own GitHub account containing the Garage Version 2 application.

Conducting the Code Review

Each student invites their group members as collaborators to their own repository. Otherwise, they will not be able to see the code or leave review comments.

Each student must create a README file containing:

- A brief description of the garage.
- How to start the program.
- Which classes exist and what each class is responsible for.
- Known limitations or areas where the student would like feedback.

This gives each reviewer context before they read the code, making the reviews more meaningful.

Git workflow

Each student should do the following in their own repository:

- Create a branch: `git checkout -b review/garage-v2`
- Create or edit the existing `README.md`.
- `git add .`
- `git commit -m "Add README for code review"`
- `git push -u origin review/garage-v2`

The PR now contains at least the README change, and group members can comment on both the README and the entire codebase through “Files changed.” `git add` stages changes, and `git commit` saves them in the branch history.

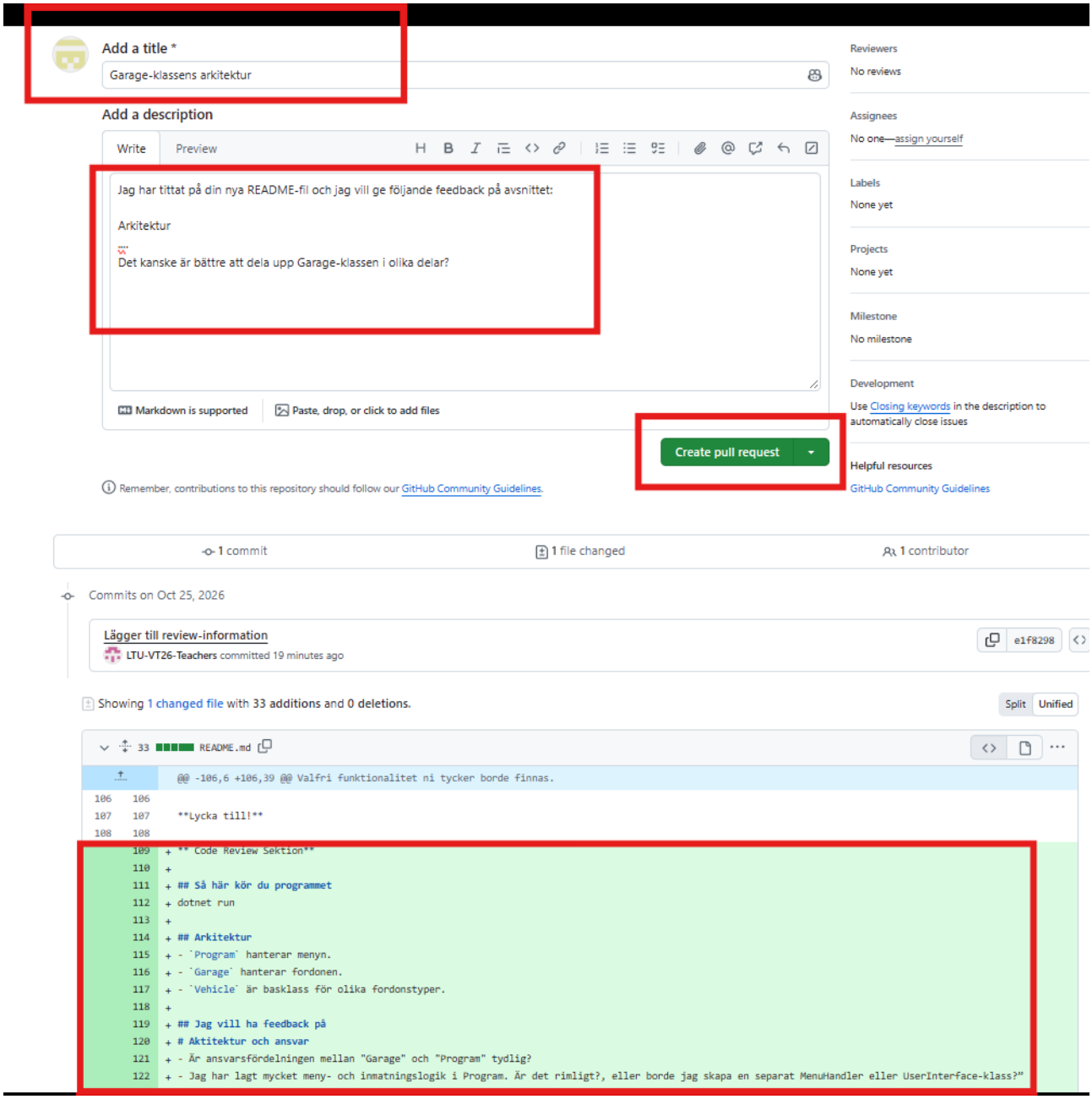
Create a Pull Request

In the `review/garage-v2` branch on GitHub, the following yellow message appears after a successful push.



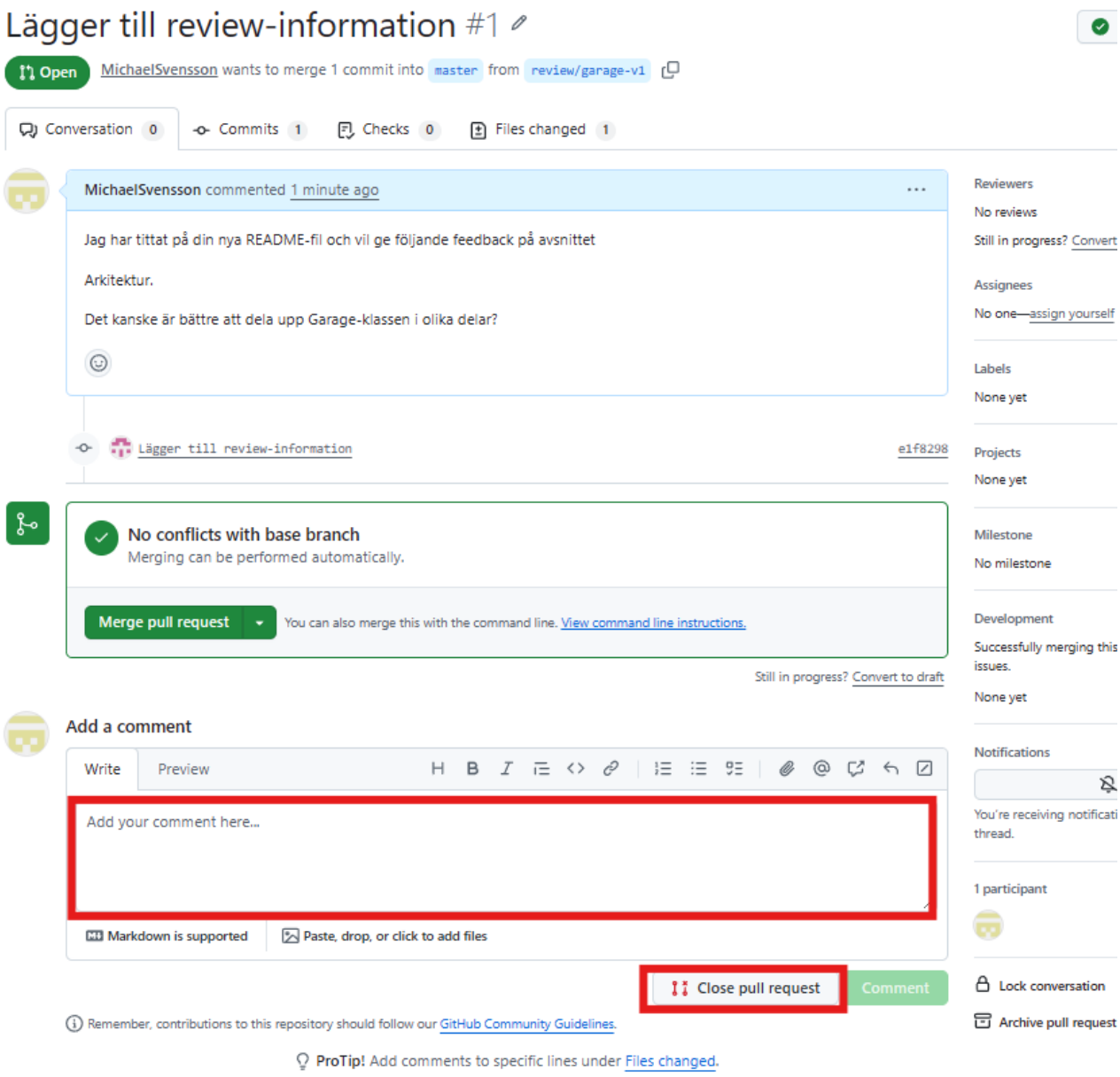
Click the “Compare & pull request” button.

Open the pull request



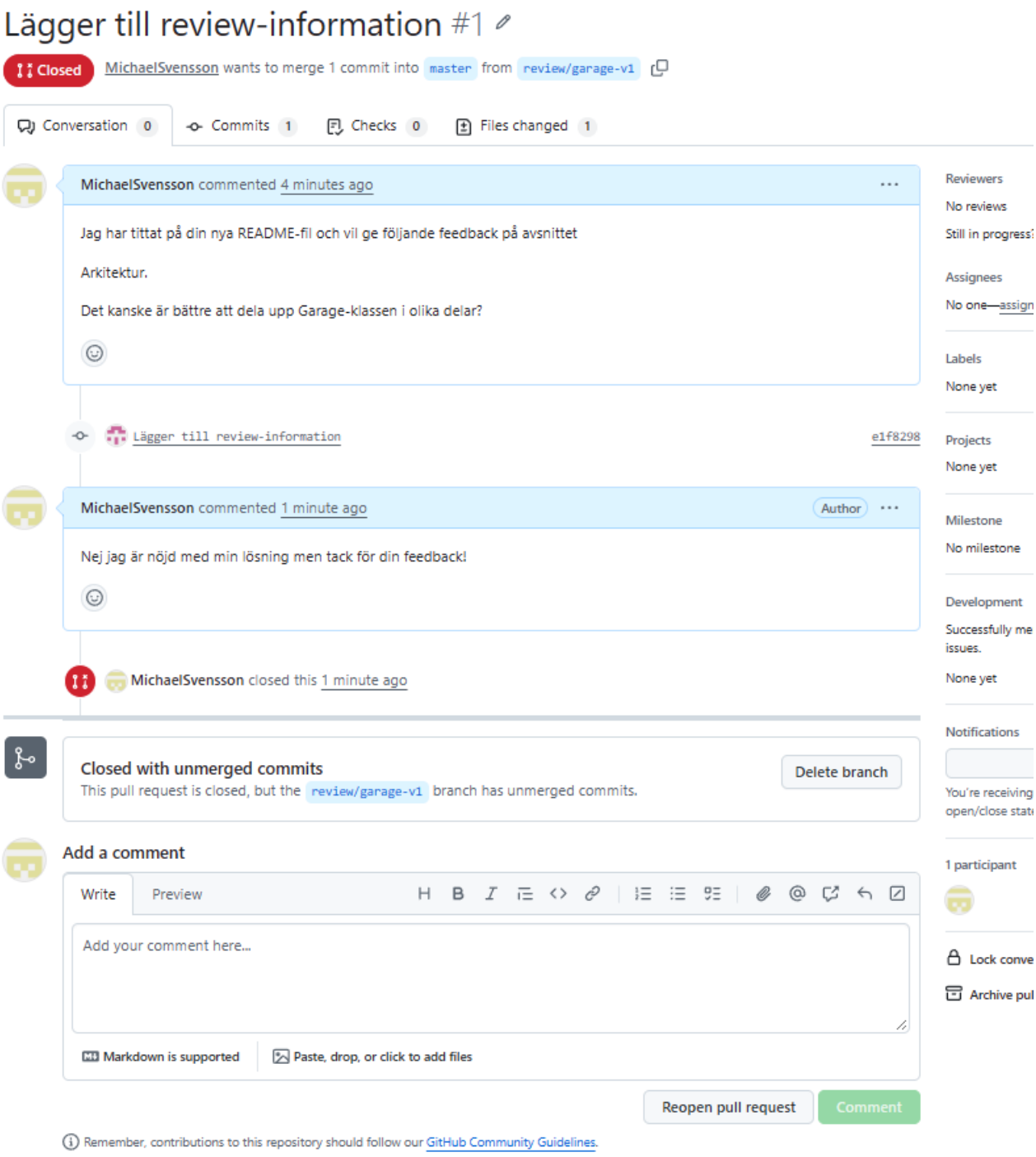
After entering a comment, as shown in the example above, click “Create pull request.”

Add a review comment

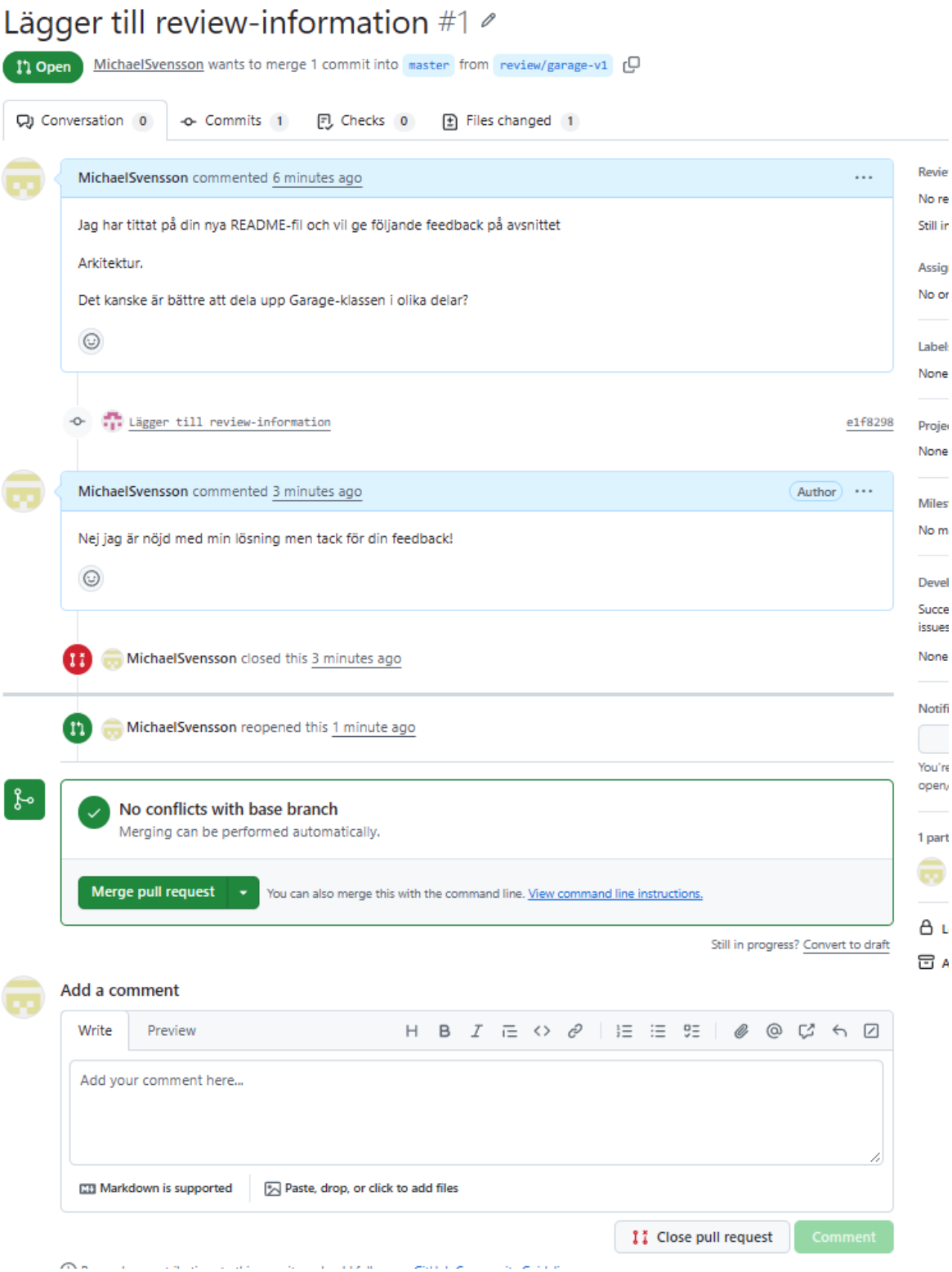


In the "Add a comment" area above, you can write a comment and then click "Comment."

Review information

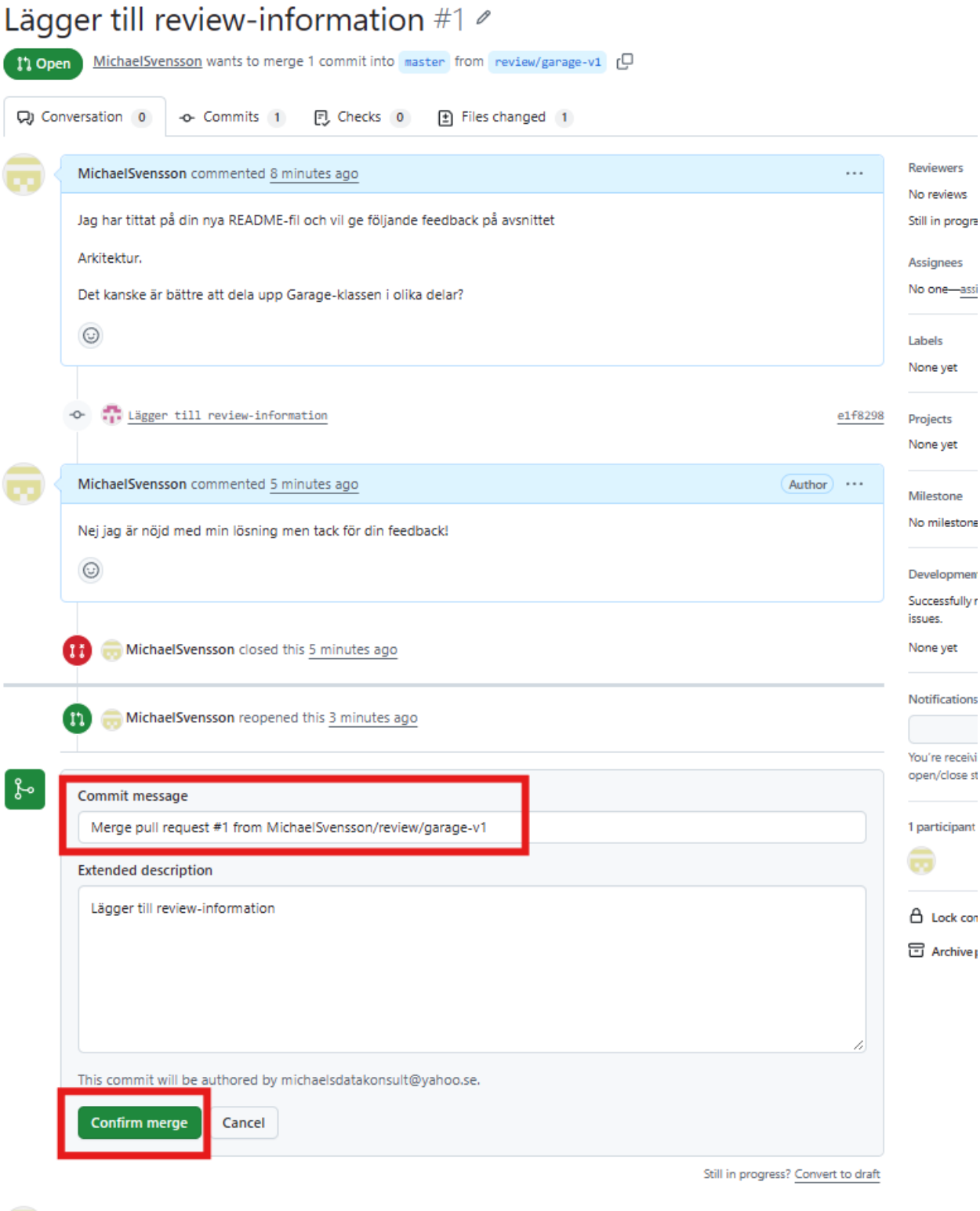


Reopen the pull request



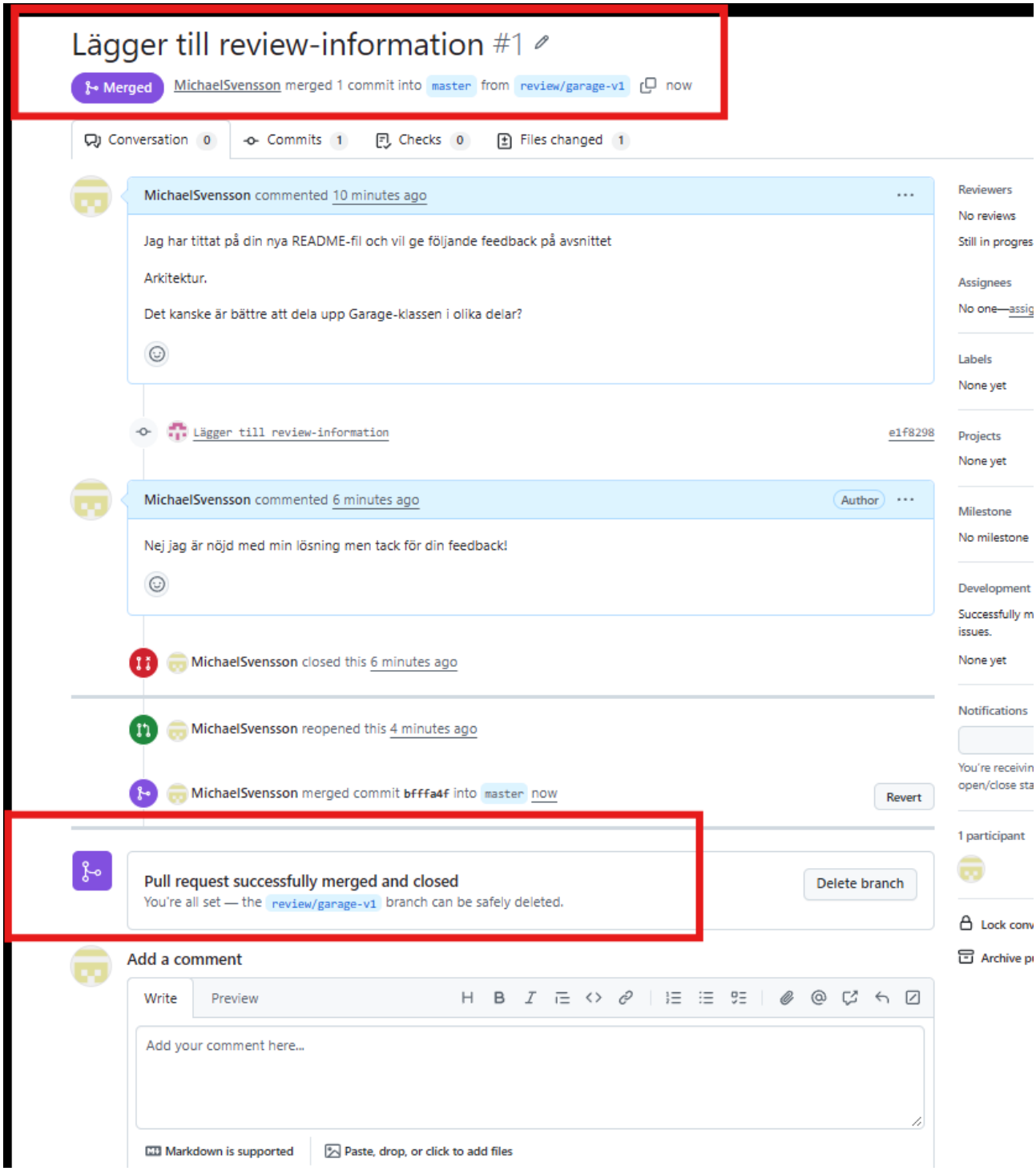
If you are satisfied with the change, click “Reopen pull request.”

Merge the pull request



Then select "Merge pull request."

Confirm the merge



Finally, click “Confirm merge.”

Create or edit the existing README.md

I would like feedback on:

Architecture and responsibilities

- Is the division of responsibilities between `Garage` and `Program` clear?
- I have put a lot of menu and input-handling logic in `Program`. Is that reasonable, or should I create a separate `MenuHandler` or `UserInterface` class?
- How can I improve handling of invalid input?
- My `Garage` class handles vehicles, capacity, searching, and printing. Does it have too many responsibilities?
- Would you split `Program.cs` into several classes? Which classes would you create, and why?

OOP and class design

- Is it clear which properties should be in the base class `Vehicle` and which should belong in subclasses?
- How should I handle capacity when different vehicles take up different amounts of space—for example, a motorcycle takes one-third of a space and a boat takes two spaces?

Garage and List

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Searching and filtering

- I search by registration number, colour, and number of wheels separately. How can I build a search with several optional criteria?

Error handling and robustness

- How should the program behave if someone tries to park a vehicle with a registration number that already exists?
- What happens if the garage is full and the user tries to park a vehicle?

Group Assignments

Group Exercise – Code Review of Garage V2.0

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